Events - Handling, Bubbling & Event Delegation in JavaScript Assignment

PW EARNERS : Coding and Web Development

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LEC - 40

Topics Covered

- addEventListener()
- Event Object and its properties
- removeEventListener()
- once option
- Event Bubbling vs Capturing
- stopPropagation()
- preventDefault()
- Event Delegation Pattern

Section 1 - Event Handling Basics

1. Handle a Button Click

Create a button and use addEventListener() to display a message when the button is clicked.

Example:
HTML:
<button id="btn">Click Me</button>
Output:
Button Clicked!

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2. Change Text on Click

Create a paragraph and a button. Use addEventListener() to change the paragraph text when the button is clicked.

Example:
Before:
Welcome to my website!
After clicking the button:
Thanks for visiting!

---

3. Handle a Mouseover Event

Create a heading and use addEventListener() with the mouseover event to change its text when the mouse moves over it.

Example:
Before:
JavaScript
After mouseover:
Mouse is over the heading!

Section 2 - Event Object

4. Display the Clicked Element

Create a button and use the event object to identify the element that was clicked.

Example:
HTML:
<button id="btn">Click Me</button>
Output:
BUTTON

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5. Display Mouse Coordinates

Create a and use a mousemove event to display the mouse coordinates using the event object's clientX and clientY properties.

Example Output:
X: 250
Y: 180

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6. Get the Value of an Input Using the Event Object

Create an input field and use the event object's target.value to display the entered value.

Example:
Input:
JavaScript
Output:
You typed: JavaScript

Section 3 - Removing and Controlling Events

7. Remove an Event Listener

Create a button and attach a click event using addEventListener(). Create a separate function and use removeEventListener() to stop the click event when required.

Example:
Before removing the listener:
Button clicked!
After removing the listener:
Clicking the button should no longer display the message.

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8. Run an Event Only Once

Create a button and use addEventListener() with the once option so that the event runs only the first time the button is clicked.

Example:
First click:
Welcome!
Second click:
No message should be displayed.

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9. Stop Event Propagation

Create a parent containing a button. Add click events to both parent and button. Use stopPropagation() so that clicking the button does not trigger the parent's click event.

Example:
Without stopPropagation():
Button clicked
Parent clicked
With stopPropagation():
Button clicked

Section 4 - Bubbling, Capturing & Default Actions

10. Demonstrate Event Bubbling

Create a parent containing a child <button>. Add click events to both elements and observe the order in which the events execute when the button is clicked.

Example:
HTML Structure:

```
<div id="parent">
  <button id="child">Click Me</button>
</div>
```

Expected Output:
Button clicked
Parent clicked

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11. Demonstrate Event Capturing

Create a parent containing a button. Add click event listeners to both using the capturing phase and observe the order in which the events execute.

Example:
Expected Output:
Parent clicked
Button clicked

Section 5 - Event Delegation

12. Handle Multiple Buttons Using Event Delegation

Create multiple buttons inside a parent . Add only one event listener to the parent and use event delegation to identify which button was clicked.

Example:
HTML:

```
<div id="buttons">
  <button>HTML</button>
  <button>CSS</button>
  <button>JavaScript</button>
</div>
```

When JavaScript button is clicked:
JavaScript button clicked

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13. Handle a Dynamic List Using Event Delegation

Create a list of items and add one event listener to the <ul>. Use event delegation to display the text of the clicked list item.

Example:
HTML:

```
<ul id="skills">
  <li>HTML</li>
  <li>CSS</li>
  <li>JavaScript</li>
</ul>
```

When CSS is clicked:
You clicked: CSS